Malakala Bala Subrahmanya Aravind

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.06
LAKSHYA SCHOOL	Class XII (CBSE)	2024	92.80%
ADITYA HIGH SCHOOL	Class X (Andhra pradesh state board(SSC))	2022	94.83%

RELEVANT COURSES

Core courses : SDF | Analog electronics | Digital systems | Signal Processing | Control systems | Circuit and network analysis

Math courses : Differential equations and transform techniques | Probability and Random processes | Machine Learning

Programming: Introduction to programming | DSA | scientific programming for Electrical Engineers

Lab Courses : Electrical Circuits Lab | Electronic Devices and Circuits Lab | Digital systems Lab | DSP Lab | PCB Design

SKILLS

Programming Languages: C | Python | MATLAB | Verilog HDL | MySQL

Development Tools : Linux | Git | LaTex

Software and simulation Tools : LTspice | Altium Designer | Arduino IDE | STM32cubeide| Icarus verilog | Intel Quartus Prime

PROJECTS

Audio Fingerprinting -- DSP / Music Information Retrieval |(Ongoing)

- Developing a **Shazam-inspired audio identification system** using STFT-based spectral analysis and spectral peak detection.
- Implementing **time-frequency feature extraction and spectral peak pairing** to generate compact audio fingerprints.
- Building a **SQLite-based fingerprint database** for storing fingerprints, song identifiers, and time offsets.
- Implementing **time-offset based matching** to identify songs from short audio clips.
- Implemented **time-offset based matching** to identify songs from short audio clips starting at arbitrary positions.

Signal Processing Algorithms using Python

- Implemented signal synthesis, **sampling**, **quantization**, and aliasing analysis using **NumPy** and **Matplotlib**.
- Developed **LTI system** simulations, including **linear convolution**, impulse response analysis, and moving-average filtering.
- Implemented **DFT**, **IDFT**, **DFT matrix**, and **8-point FFT**, validating results against NumPy implementations.
- Verified DFT properties, performed **frequency-domain filtering**, and analyzed magnitude and phase spectra **of signals**.
- **Processed** audio and speech signals through sampling, quantization, filtering, and spectral analysis using **Python**.

Fixed-Point DSP Architectures using Verilog HDL

- Developed **MATLAB** scripts for FIR coefficient generation, fixed-point signal synthesis, and DSP algorithm verification.
- Implemented **100-tap FIR filters**, **8-point Radix-2 FFT**, and **Newton-Raphson** fixed-point division in **Verilog HDL**.

- Designed DSP hardware using **pipelining**, **coefficient symmetry**, and **LUT-based** Newton-Raphson initialization.
- Verified Verilog designs against **MATLAB** models and **Intel FFT IP**, analyzing **Fmax**, throughput, and resources.

STM32-Based Diode V-I Characterization System

- Designed a **4-layer PCB** using **STM32F303C8T6** integrating **12-bit DAC**, **12-bit ADC**, **USB-UART**, and power management.
- Implemented embedded firmware in **STM32CubeIDE** for **DAC sweeping**, synchronized **ADC sampling**, and data acquisition.
- Designed the analog front-end using a **1.65 V reference**, **MCP6002 op-amp**, **current-sensing resistor**, and diode interface.
- Integrated **CP2102 USB-UART** for real-time transfer of voltage and current measurements to a PC.
- Validated the measurement system through **DAC/ADC calibration**, hardware testing, and experimental verification.
- Generated forward and reverse **diode V-I characteristics** by processing experimental data in **Python**.

5-Stage Pipelined RV64I RISC-V Processor using Verilog HDL

- Designed & implemented a **64-bit RV64I RISC-V processor**, evolving from a single-cycle to a **5-stage pipelined architecture**.
- Developed the **Program Counter**, **Register File**, **ALU**, **Control Unit**, **Immediate Generator**, and pipeline registers.
- **Implemented Instruction Memory**, **Data Memory**, and datapath control across all **five pipeline stages**.
- Designed a **Forwarding Unit** with forwarding multiplexers to resolve **ALU data hazards** and improve pipeline throughput.
- Verified processor functionality using **Verilog testbenches** by executing RV64I assembly programs and debugging waveforms.
- Validated arithmetic operations, branching, memory access, and complete pipelined execution across multiple test programs.

POSITIONS OF RESPONSIBILITY

Events coordinator | SPICMACAY | 2025-2026

- Coordinated the planning and execution of SPIC MACAY cultural events.
- Assisted with event logistics and on-ground operations.

EXTRACURRICULAR

Active hockey player; represented the **EE branch** in **Diesta** and the **hostel** in **Milan** and played **IHL**

Worked as a volunteer for **Elan**, assisting with on-ground event execution and logistics.